

Equilibrium

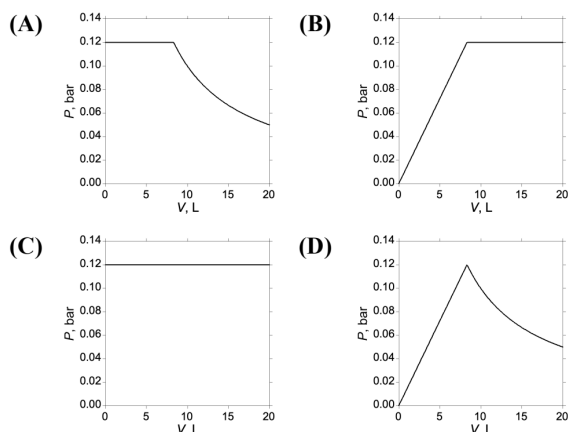
1. How much $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ must be added to 400.0 mL of 0.200 M H_3PO_4 to create a buffer solution with $\text{pH} = 2.30$? For H_3PO_4 , $\text{p}K_{\text{a}1} = 2.12$, $\text{p}K_{\text{a}2} = 7.21$, and $\text{p}K_{\text{a}3} = 12.38$.

(A) 0.0144 mol (B) 0.0529 mol
(C) 0.121 mol (D) 0.303 mol

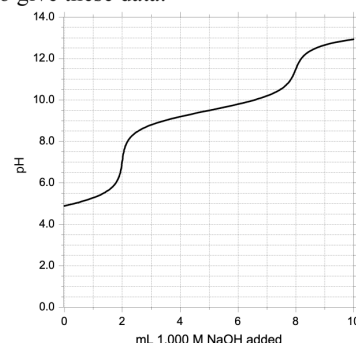
2. Gaseous PCl_5 is introduced into a container whose volume is adjustable. At equilibrium at a certain temperature, 20.0% of the PCl_5 has decomposed at equilibrium to form PCl_3 and Cl_2 . Increasing the volume of the container by what factor will increase the percentage of decomposition to 50.0% at equilibrium at this temperature?

(A) 2.50 (B) 4.00
(C) 6.25 (D) 10.0

3. Solid calcium carbonate is in equilibrium with calcium oxide and carbon dioxide, with $K_p = 0.12$ bar at 1200 K.
 $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g}) \quad K_{\text{eq}} = 0.12$ at 1200 K
A 1.00 g sample of CaCO_3 ($M = 100.09$) is placed in an evacuated piston which is allowed to equilibrate at 1200 K. How will the pressure in the piston after equilibrium is attained depend on the volume of the piston?



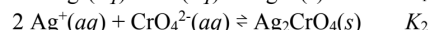
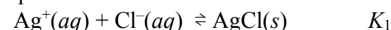
4. A mixture of the diprotic acid $\text{HOC}_6\text{H}_4\text{COOH}$ and $\text{Na}[\text{HOC}_6\text{H}_4\text{COO}]$ is titrated with sodium hydroxide solution to give these data:



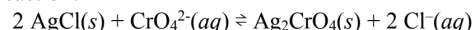
Which statement is correct?

- (A) The ratio of $\text{Na}[\text{HOC}_6\text{H}_4\text{COO}]$ to $\text{HOC}_6\text{H}_4\text{COOH}$ is 2:1.
(B) The total number of moles of $\text{Na}[\text{HOC}_6\text{H}_4\text{COO}]$ and $\text{HOC}_6\text{H}_4\text{COOH}$ combined is 0.008.
(C) The $\text{p}K_{\text{a}1}$ of $\text{HOC}_6\text{H}_4\text{COOH}$ is 5.3.
(D) The $\text{p}K_{\text{a}2}$ of $\text{HOC}_6\text{H}_4\text{COOH}$ is 9.2.
5. 0.50 mol of solid silver chloride ($K_{\text{sp}} = 1.8 \times 10^{-10}$) is added to 1.0 L of a 0.100 M solution of sodium bromide. What is the concentration of silver ion in solution at equilibrium? The K_{sp} of AgBr is 4.9×10^{-13} .
- (A) 4.9×10^{-12} M (B) 1.8×10^{-9} M
(C) 7.0×10^{-7} M (D) 1.3×10^{-5} M
6. A solution initially is 0.10 M in both Cd^{2+} and Tl^+ and is kept saturated with hydrogen sulfide gas ($[\text{H}_2\text{S}] = 0.1$ M). In what pH range will one of the metal ions be precipitated quantitatively ($> 99.9\%$) while the other remains completely in solution?
- K_{sp} of $\text{CdS} = 1.0 \times 10^{-27}$ K_{sp} of $\text{Tl}_2\text{S} = 6.0 \times 10^{-22}$
 K_{a} of $\text{H}_2\text{S} = 8.9 \times 10^{-8}$ K_{a} of $\text{HS}^- = 1.0 \times 10^{-19}$
- (A) Between 0.5 and 6.8
(B) Between 2.0 and 3.9
(C) Between 4.0 and 6.8
(D) There is no pH at which this is possible.

7. Given the equilibrium constants:



What is the equilibrium constant for the following reaction?



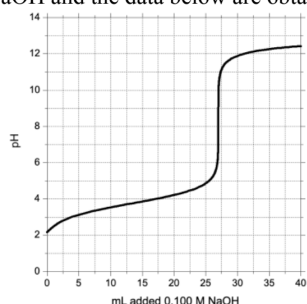
(A) $\frac{K_1}{K_2}$ (B) $\frac{K_1}{K_2^2}$
(C) $\frac{K_2^2}{K_1}$ (D) $\frac{K_2}{K_1^2}$

Equilibrium

8. A solution of allylamine, $\text{C}_3\text{H}_5\text{NH}_2$, is 3.61% ionized and has a pH = 11.12. What is the $\text{p}K_a$ of the allylammonium ion, $\text{C}_3\text{H}_5\text{NH}_3^+$?

(A) 3.77 (B) 9.69
(C) 10.23 (D) 12.56

9. A solution of formic acid, HCOOH , is titrated with 0.1000 M NaOH and the data below are obtained.



At the point where 15.0 mL of NaOH have been added, what are the relative concentrations of HCOOH , HCOO^- , and Na^+ ?

(A) $[\text{HCOO}^-] > [\text{Na}^+] > [\text{HCOOH}]$
(B) $[\text{Na}^+] > [\text{HCOO}^-] > [\text{HCOOH}]$
(C) $[\text{HCOOH}] > [\text{Na}^+] > [\text{HCOO}^-]$
(D) $[\text{HCOOH}] > [\text{HCOO}^-] > [\text{Na}^+]$

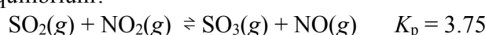
10. At high temperatures, solid calcium carbonate ($M = 100.09$) is in equilibrium with solid calcium oxide and carbon dioxide gas.



Into an evacuated 30.0 L chamber maintained at 1200 K is introduced 3.0 g solid CaCO_3 . After the system is allowed sufficient time to equilibrate, the pressure is 0.10 bar. What will the total pressure be at equilibrium if an additional 3.0 g CaCO_3 is added to the chamber?

(A) 0.10 bar
(B) 0.14 bar
(C) 0.20 bar
(D) It cannot be determined from the information given.

11. At a certain temperature, the reaction below has $K_p = 3.75$. Initially, all four gases are in a rigid container, each with a partial pressure of 0.60 bar. What will the partial pressure of $\text{SO}_2(g)$ be once the system achieves equilibrium?



(A) 0.19 bar (B) 0.25 bar
(C) 0.35 bar (D) 0.41 bar

12. Cadmium iodate, $\text{Cd}(\text{IO}_3)_2$, has $K_{sp} = 2.5 \times 10^{-8}$. Iodide ion forms a complex ion CdI_4^{2-} with $K_f = 1.3 \times 10^6$. What is the minimum amount of iodide ion that would need to be added to dissolve 0.100 mol $\text{Cd}(\text{IO}_3)_2$ in 1.00 L of solution?

(A) 0.40 mol (B) 0.59 mol
(C) 0.62 mol (D) 0.99 mol

13. A sample of 0.320 mol of a weak monoprotic acid in 1.00 L of solution is 10.2% ionized. What is K_a for this acid?

(A) 3.3×10^{-3} (B) 3.7×10^{-3}
(C) 3.2×10^{-2} (D) 1.1×10^{-1}

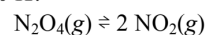
14. Typically, the molar solubility x of an ionic compound in pure water and its K_{sp} are related by a simple formula. Which formula is not expected for K_{sp} for an ionic compound?

(A) $K_{sp} = 4x^3$ (B) $K_{sp} = 4x^4$
(C) $K_{sp} = 27x^4$ (D) $K_{sp} = 81x^5$

15. Equal volumes of 0.1 M solutions of NaH_2PO_4 and Na_3PO_4 are mixed. What is the pH of the resulting solution? For H_3PO_4 , $\text{p}K_{a1} = 2.12$, $\text{p}K_{a2} = 7.21$, $\text{p}K_{a3} = 12.32$.

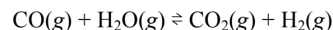
(A) 4.66 (B) 6.66 (C) 7.21 (D) 9.76

16. 5.00 g of nitrogen dioxide is introduced into an evacuated, rigid 1.00 L container. It is allowed to equilibrate according to the reaction below at 310 K, and the total pressure in the vessel is 1.71 bar. What is K_p for the reaction at 310 K?



(A) 0.35 (B) 0.57 (C) 2.28 (D) 2.92

17. A mixture containing $\text{CO}(g)$, $\text{H}_2\text{O}(g)$, $\text{CO}_2(g)$, and $\text{H}_2(g)$ is allowed to reach equilibrium at 120 °C in a container that maintains a constant total pressure of 1.0 bar.



5.0 g of H_2O is added to this container, which is allowed to reattain equilibrium (still at 120 °C and 1 bar total pressure). How does the partial pressure of $\text{H}_2(g)$ change as a result of the addition of H_2O ?

(A) The partial pressure of $\text{H}_2(g)$ increases.
(B) The partial pressure of $\text{H}_2(g)$ decreases.
(C) The partial pressure of $\text{H}_2(g)$ does not change.
(D) The effect on the partial pressure of $\text{H}_2(g)$ cannot be determined from the information given.

Equilibrium

18. 0.01 mol of AgNO_3 is dissolved in 1.00 L of 1.00 M aqueous NH_3 . What is the minimum amount of HCl that would need to be added to this solution to induce precipitation of AgCl ? The K_{sp} of AgCl is 1.8×10^{-10} ; the K_{f} of $\text{Ag}(\text{NH}_3)_2^+$ is 1.6×10^7 ; and the K_{a} of NH_4^+ is 5.6×10^{-10} .

(A) 1.8×10^{-8} mol (B) 1.3×10^{-5} mol
(C) 0.18 mol (D) 0.28 mol

19. Pure PCl_5 is placed in a rigid flask so that its pressure at 250 °C is 2.000 bar. The gas slowly decomposes to PCl_3 and Cl_2 at this temperature, and the partial pressure of Cl_2 is eventually 0.814 bar. What is the value of K_{p} for the reaction?

$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \quad K_{\text{p}} = ???$
(A) 0.559 (B) 0.686 (C) 0.814 (D) 1.78

20. What is the maximum concentration of lead(II) ions at equilibrium in a solution with $\text{pH} = 10.50$? The K_{sp} of $\text{Pb}(\text{OH})_2$ is 1.4×10^{-15} .

(A) 7.1×10^{-6} M (B) 1.4×10^{-8} M
(C) 3.5×10^{-9} M (D) 4.4×10^{-12} M

21. The $\text{p}K_{\text{a}}$ of methylammonium ion, CH_3NH_3^+ , is 10.64. What is the percent ionization of a 0.20 M solution of methylamine, CH_3NH_2 ?

(A) 0.0011% (B) 0.071%
(C) 1.1% (D) 4.7%

22. Which statement best describes the differences between a 0.1 M solution of ammonium bicarbonate, $\text{NH}_4(\text{HCO}_3)$, and a 0.1 M solution of ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$?

(A) The pH of the ammonium bicarbonate solution is lower because bicarbonate is a weaker base than carbonate.
(B) The pH of the ammonium bicarbonate solution is lower because both ammonium ion and bicarbonate ion can act as Brønsted acids.
(C) The pH of the ammonium bicarbonate solution is higher because it has only half the ammonium ion concentration of the ammonium carbonate solution.
(D) The pH of the ammonium bicarbonate solution is higher because it contains only two-thirds as many total ions as the ammonium carbonate solution.

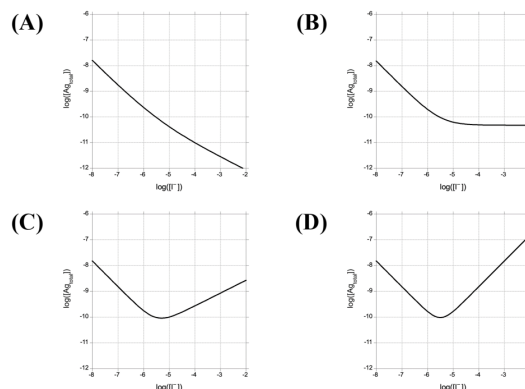
23. A mixture of $\text{N}_2\text{O}_4(\text{g})$ and $\text{NO}_2(\text{g})$ is contained in a vessel whose total volume can change but where the total pressure is maintained at 1.00 atm. The initial temperature is 298 K, and the two gases are in equilibrium. Which changes will result in an increase in the number of moles of NO_2 present at equilibrium?



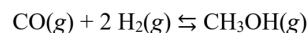
I. Adding $\text{Ne}(\text{g})$
II. Increasing the temperature to 310 K

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

24. Silver iodide is a sparingly soluble salt. Silver also forms a soluble complex ion, AgI_2^- , with iodide ion. A series of solutions saturated with solid AgI and containing various concentrations of dissolved iodide ion are prepared, and the total concentration of silver dissolved in each solution is measured. Which graph of the logarithm of the total silver concentration as a function of the logarithm of the iodide concentration best represents the results of this experiment?



25. Carbon monoxide reacts with hydrogen in the presence of a catalyst to give methanol:



A metal container at 400 K is charged with 2.00 bar of an equimolar mixture of CO and H_2 . A catalyst for the reaction is introduced and the pressure falls to 1.29 bar once equilibrium is achieved. What is K_{p} for this reaction at 400 K?

(A) 1.32 (B) 6.54 (C) 8.44 (D) 29.1

26. A 0.10 M solution of glycolic acid is 3.8% ionized. What is the percent ionization of a 0.50 M solution of glycolic acid?

(A) 1.7% (B) 3.8% (C) 8.5% (D) 19.0%

Equilibrium

27. The K_{sp} of BaCrO_4 is 1.2×10^{-10} . What is the minimum volume of $1.5 \times 10^{-5} \text{ M}$ Na_2CrO_4 that would need to be added to 1.0 L of $1.5 \times 10^{-5} \text{ M}$ $\text{Ba}(\text{NO}_3)_2$ solution to induce precipitation of BaCrO_4 ?

(A) 0.53 L
(B) 0.73 L
(C) 1.0 L

(D) No amount of this Na_2CrO_4 solution will cause precipitation of BaCrO_4 .

28. A reaction has $K_{eq} = 0.020$ at 300 K, and its K_{eq} value increases with increasing temperature. What can be inferred about the values of ΔH°_{rxn} and ΔS°_{rxn} , assuming that they are independent of temperature?

I. $\Delta H^\circ_{rxn} < 0$

II. $\Delta S^\circ_{rxn} < 0$

(A) I only

(B) II only

(C) Both I and II

(D) Neither I nor II

29. Silver ion forms the complex $\text{Ag}(\text{CN})_2^-$ with $K_f = 9.8 \times 10^{21}$. What is the minimum amount of HCN that would need to be added to 1.00 L of a suspension of 0.010 mol AgCl in order to dissolve all the solid? The K_{sp} of AgCl is 1.8×10^{-10} and the K_a of HCN is 6.2×10^{-10} .

(A) 0.020 mol

(B) 0.079 mol

(C) 0.26 mol

(D) 1.7 mol

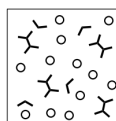
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30. A vessel containing an equilibrium mixture of $\text{NO}_2(\text{g})$ and $\text{N}_2\text{O}_4(\text{g})$ at a certain temperature and 1.00 bar total pressure is represented below.

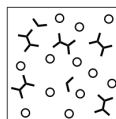


Which is the best representation of the system at equilibrium after a certain amount of argon is added to the vessel, while keeping both the temperature and the total pressure constant?

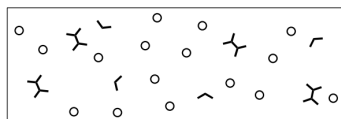
(A)



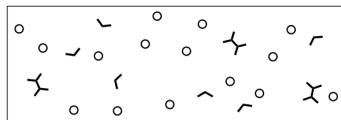
(B)



(C)



(D)



31. What is the percent ionization of a 0.10 M solution of nitrous acid (HNO_2 , $K_a = 7.2 \times 10^{-4}$)?

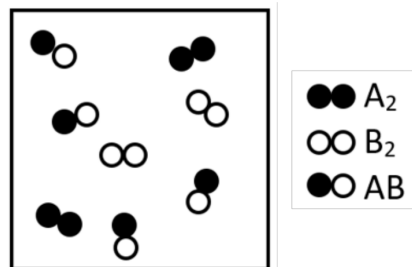
(A) 1.1%

(B) 3.6%

(C) 8.1%

(D) 14%

32. The reaction mixture below represents the reaction $\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons 2 \text{AB}(\text{g})$ with $K_{eq} = 2.5$. Which statement correctly describes the situation?



(A) The reaction is at equilibrium.

(B) The reaction will shift to produce more products.

(C) The reaction will shift to produce more reactants.

(D) Not enough information is given to make a prediction.

Equilibrium

33. The hydrogenation of carbon monoxide represented below is exothermic.



Which changes will increase the equilibrium yield of CH_3OH ?

- I. Increasing the temperature of the system
- II. Removing some of the $\text{CH}_3\text{OH(l)}$

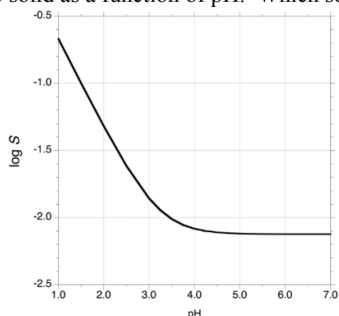
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

34. To a solution that is 0.010 M each in fluoride, sulfite, and phosphate ions is added dropwise a 1.00 M solution of CaCl_2 . In what order will the solids precipitate?

Solid	K_{sp}
CaSO_3	6.8×10^{-8}
CaF_2	5.3×10^{-9}
$\text{Ca}_3(\text{PO}_4)_2$	1.0×10^{-25}

- (A) First $\text{Ca}_3(\text{PO}_4)_2$, then CaF_2 , last CaSO_3
(B) First $\text{Ca}_3(\text{PO}_4)_2$, then CaSO_3 , last CaF_2
(C) First CaSO_3 , then CaF_2 , last $\text{Ca}_3(\text{PO}_4)_2$
(D) First CaF_2 , then CaSO_3 , last $\text{Ca}_3(\text{PO}_4)_2$

35. The graph shows the logarithm of the molar solubility S of an ionic solid as a function of pH. Which solid is it?

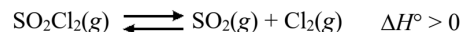


- (A) $\text{Ag}_2\text{C}_2\text{O}_4$ (B) AlPO_4
(C) SrSO_4 (D) BaF_2

36. Some solid silver nitrate is dissolved in 100.0 mL of 1.00 M NH_3 solution. To this solution, 50.0 mL of 1.00 M NaCl solution must be added to see the first sign of cloudiness. How much AgNO_3 was present? The K_{sp} of AgCl is 1.8×10^{-10} and the K_f of $\text{Ag}(\text{NH}_3)_2^+$ is 1.0×10^8 .

- (A) 8.1×10^{-11} mol (B) 2.1×10^{-3} mol
(C) 3.2×10^{-3} mol (D) 4.9×10^{-3} mol

37. A sample of SO_2Cl_2 is introduced into a movable piston with the pressure maintained at 1.00 atm and the temperature at 450 K. The volume increases as the sulfuryl chloride equilibrates according to the following endothermic reaction:



Which changes would decrease the number of moles of $\text{SO}_2\text{Cl}_2\text{(g)}$ present in the piston at equilibrium? In each case, the pressure is maintained at 1.00 atm.

- I. The temperature of the system is increased to 500 K.
- II. A sample of Ar(g) is injected into the piston.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

38. Equal volumes of 0.500 M HCO_2H and 0.100 M $\text{Na}(\text{O}_2\text{CCH}_3)$ are mixed. What is the pH of the resulting solution? The $\text{p}K_a$ of HCO_2H is 3.75 and the $\text{p}K_a$ of $\text{CH}_3\text{CO}_2\text{H}$ is 4.75.

- (A) 2.18 (B) 3.15 (C) 3.75 (D) 4.25

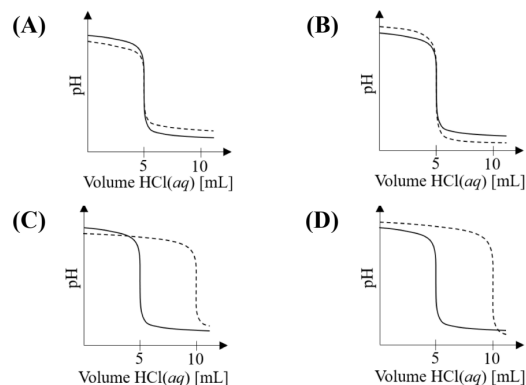
39. The formation constant K_f of the hexamminecadmium ion, $\text{Cd}(\text{NH}_3)_6^{2+}$, is 2.6×10^5 . In a solution whose ammonia concentration is 0.12 M, what percentage of the cadmium in solution is in the form of $\text{Cd}(\text{NH}_3)_6^{2+}$?

- (A) 44% (B) 51% (C) 78% (D) > 99%

40. The K_{sp} of $\text{Sr}(\text{OH})_2$ is 2.0×10^{-3} . What is the pH of a saturated solution of $\text{Sr}(\text{OH})_2$?

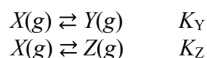
- (A) 11.30 (B) 12.65 (C) 12.90 (D) 13.20

41. When 0.100 M HCl was added to a 10.0 mL sample of aqueous NaOH , it generated the titration curve shown as a solid line. If the same number of moles of NaOH , but initially at a volume of 20.0 mL, were titrated with 0.100 M HCl , which of the dashed curves would best describe this new titration?

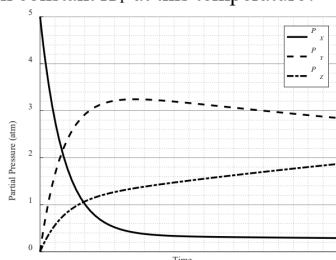


Equilibrium

42. A gas X reacts reversibly and independently to form two different gases, Y and Z :

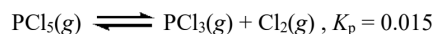


At a certain temperature, the time evolution of the pressures of the three gases are shown. What is the equilibrium constant K_Y at this temperature?

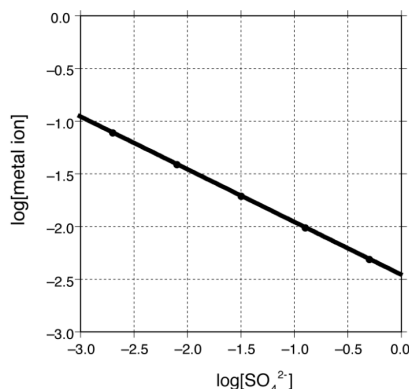


- (A) 3.0
(B) 8.0
(C) 9.3
(D) It cannot be determined because the system has not yet reached equilibrium.
43. A 0.1 M solution of which compound is most acidic?
(A) KNO_3 (B) NH_4NO_3
(C) $\text{Ba}(\text{NO}_3)_2$ (D) $\text{Fe}(\text{NO}_3)_3$
44. The molar solubility of PbF_2 is $2.1 \times 10^{-3} \text{ mol L}^{-1}$. What is its K_{sp} ?
(A) 4.4×10^{-6} (B) 8.8×10^{-6}
(C) 3.7×10^{-8} (D) 9.3×10^{-9}
45. What is the pH of a 0.10 M solution of ammonium acetate, $\text{NH}_4(\text{CH}_3\text{COO})$? The K_a of NH_4^+ is 5.6×10^{-10} and the K_a of CH_3COOH is 1.8×10^{-5} .
(A) 2.87 (B) 5.13 (C) 7.00 (D) 8.87

Problems 46 and 47 both concern the dissociation of phosphorus pentachloride into chlorine and phosphorus trichloride, which has $K_p = 0.015$ at 450 K.



46. A closed container at 450 K initially contains only $\text{PCl}_5(g)$ and $\text{PCl}_3(g)$, each with a partial pressure of 2.7 bar. After the system reaches equilibrium, what is the partial pressure of $\text{Cl}_2(g)$?
(A) 0.015 bar (B) 0.12 bar
(C) 0.20 bar (D) 2.7 bar
47. Which change will decrease the number of moles of $\text{Cl}_2(g)$ present in this system at equilibrium?
(A) Increasing the volume of the container.
(B) Increasing the pressure by injecting $\text{Ar}(g)$.
(C) Increasing the pressure by injecting $\text{PCl}_5(g)$.
(D) Decreasing the temperature.
48. A metal M forms a sparingly soluble sulfate salt. The logarithm of the concentration of the dissolved metal ion in a solution saturated in the metal sulfate is plotted below as a function of the logarithm of the concentration of sulfate ion in solution. What is the formula of the metal sulfate?



- (A) M_2SO_4 (B) MSO_4
(C) $\text{M}_2(\text{SO}_4)_3$ (D) $\text{M}(\text{SO}_4)_2$

Equilibrium

49. A 1.00 g pill contains morphine ($M = 285.34$), a monobasic compound whose conjugate acid has $pK_a = 8.21$, in addition to an unknown amount of inert material. To analyze the morphine content of the pill, it is dissolved in 50.0 mL 0.1000 M aqueous HCl and the unreacted acid in the resulting solution back-titrated with 0.1000 M NaOH. Which of the following statements about this experiment are correct?
- If the pill contains only morphine, the titration will require 35.0 mL NaOH solution to reach the endpoint.
 - Methyl red, which changes color from red to yellow between $pH = 4.4$ and $pH = 6.2$, would be a suitable indicator for this titration.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
50. Which species is the strongest Brønsted acid?
- (A) HBrO (B) HBrO₂
(C) HBrO₃ (D) HBrO₄
51. What is the pH of a saturated solution of $Co(OH)_2$? The K_{sp} of $Co(OH)_2$ is 5.9×10^{-15} .
- (A) 7.19 (B) 9.06 (C) 9.36 (D) 9.56
52. Which of the following solutions is capable of dissolving the most $CaC_2O_4(s)$?
- (A) Pure H₂O (B) 0.10 M $CaCl_2$
(C) 0.10 M $H_2C_2O_4$ (D) 0.10 M HCl
53. Silver ion forms a complex ion with thiosulfate ion, $Ag(S_2O_3)_2^{3-}$, with $K_f = 2.8 \times 10^{13}$. How much AgBr ($K_{sp} = 5.4 \times 10^{-13}$) will dissolve in 1.00 L of 0.200 M $Na_2S_2O_3$ solution?
- (A) 0.089 mol (B) 0.10 mol
(C) 0.16 mol (D) 0.78 mol
54. 100.0 mL 0.15 M aqueous HF ($K_a = 6.8 \times 10^{-4}$) is mixed with 125.0 mL 0.23 M NaF. What is the pH of the resulting solution?
- (A) 2.17 (B) 3.17 (C) 3.35 (D) 3.45
55. For a given reaction, a graph of $\ln(K_{eq})$ vs. $1/T$ is linear and has a positive slope. Which statements about this reaction must be correct?
- $\Delta H^\circ_{rxn} < 0$
 - $\Delta S^\circ_{rxn} > 0$
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
56. The following reaction has $K_p = 50.4$ at 448 °C.
- $$H_2(g) + I_2(g) \rightleftharpoons 2 HI(g)$$
- If a 3.00 L flask initially contains 0.0500 moles each of H_2 and I_2 , how many moles of HI are present when the contents have reached equilibrium at 448 °C?
- (A) 0.0130 mol (B) 0.0146 mol
(C) 0.0260 mol (D) 0.0780 mol
57. A 0.1 M solution of which salt is the most acidic?
- (A) $Al(NO_3)_3$ (B) $MgBr_2$
(C) $NaHCO_3$ (D) $NaHCO_2$
58. What is the pH of a 0.25 M solution of NaCN? (The pK_a of HCN is 9.21.)
- (A) 4.91 (B) 8.61 (C) 11.30 (D) 13.40
59. The autoionization constant of water at 60 °C is $K_w = 1.0 \times 10^{-13}$. Which of the following statements are correct?
- Autoionization of water is exothermic.
 - A sample of pure water at 60 °C is slightly acidic.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
60. Barium carbonate, $BaCO_3$, is stable at ambient temperatures, but decomposes to barium oxide and carbon dioxide at higher temperatures.
- $$BaCO_3(s) \rightleftharpoons BaO(s) + CO_2(g)$$
- At a certain temperature, this system is in equilibrium in a closed system and contains appreciable amounts of all three compounds. Which changes will lead to an increase in the pressure of CO_2 present at equilibrium?
- Adding more $BaCO_3(s)$
 - Increasing the volume of the container
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
61. If 0.10 mol solid NaOH is added to 1.00 L of a saturated solution of $Ca(OH)_2$ ($K_{sp} = 8.0 \times 10^{-6}$), what percentage of the calcium hydroxide will precipitate at equilibrium?
- (A) Roughly 50% (B) Roughly 75%
(C) Roughly 95% (D) Over 99%

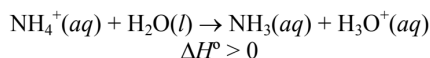
Equilibrium

62. The concentration of formic acid ($pK_a = 3.75$) is being determined by titration with sodium hydroxide solution. Which indicators are suitable for this titration?

I. Bromophenol blue (pH transition range 3.0 – 4.6)
II. Neutral red (pH transition range 6.8 – 8.0)

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

63. The ionization of ammonium ion is endothermic:

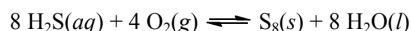


Which changes will result in the increase in $[\text{H}_3\text{O}^+]$ of a 0.100 M solution of NH_4Cl ?

I. Diluting the solution from 1.00 L to 2.00 L
II. Raising the temperature from 25 °C to 35 °C

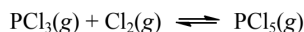
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

64. What is the correct equilibrium expression for the following reaction?



- (A) $K = \frac{[\text{S}_8][\text{H}_2\text{O}]}{[\text{H}_2\text{S}] \cdot P(\text{O}_2)}$
(B) $K = \frac{[\text{S}_8](8 \cdot [\text{H}_2\text{O}])^8}{(8 \cdot [\text{H}_2\text{S}])^8 \cdot (4 \cdot P(\text{O}_2))^4}$
(C) $K = \frac{[\text{S}_8][\text{H}_2\text{O}]^8}{[\text{H}_2\text{S}]^8 \cdot P(\text{O}_2)^4}$
(D) $K = \frac{1}{[\text{H}_2\text{S}]^8 \cdot P(\text{O}_2)^4}$

65. The reaction below is exothermic.



Which change will increase the number of moles of $\text{PCl}_5(g)$ present at equilibrium?

- (A) The volume of the reaction vessel is tripled.
(B) The reaction vessel is cooled.
(C) Some of the $\text{Cl}_2(g)$ is removed.
(D) Krypton gas is added to the reaction vessel.

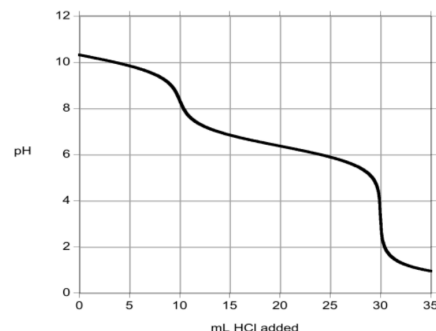
66. Which of the following has the greatest molar solubility in water?

- (A) ZnC_2O_4 ($K_{sp} = 2.7 \times 10^{-8}$)
(B) BaCrO_4 ($K_{sp} = 2.3 \times 10^{-10}$)
(C) CaF_2 ($K_{sp} = 3.9 \times 10^{-11}$)
(D) AgBr ($K_{sp} = 5.0 \times 10^{-13}$)

67. A 0.100 M solution of a weak monoprotic acid HX is 3.5% ionized. What is the pH of a 0.500 M solution of HX?

- (A) 0.30 (B) 1.76 (C) 2.10 (D) 2.46

68. A sample of a washing powder that contains a mixture of Na_2CO_3 and NaHCO_3 is titrated with aqueous HCl and the following result is obtained:



What is the mole ratio of CO_3^{2-} to HCO_3^- in the washing powder?

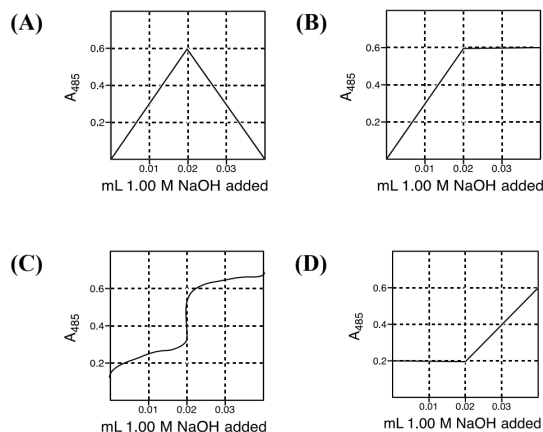
- (A) 2 mol CO_3^{2-} : 1 mol HCO_3^-
(B) 1 mol CO_3^{2-} : 1 mol HCO_3^-
(C) 1 mol CO_3^{2-} : 2 mol HCO_3^-
(D) 1 mol CO_3^{2-} : 3 mol HCO_3^-

69. Hydrocyanic acid, HCN, is a weak acid with $K_a = 4.9 \times 10^{-10}$. Nickel(II) ion complexes strongly with cyanide ion to form $\text{Ni}(\text{CN})_4^{2-}$, $K_f = 1.0 \times 10^{22}$. What is the pH of 1.00 L of a 0.100 M solution of HCN to which 0.025 mol NiCl_2 has been added?

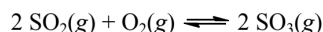
- (A) 1.00 (B) 4.05 (C) 5.15 (D) 8.92

Equilibrium

70. Nitrophenol is a colorless weak monoprotic acid ($pK_a = 7.2$) whose conjugate base is bright yellow. To 2.00 mL of a solution of 0.0100 M nitrophenol is added 1.00 M NaOH in 0.001 mL portions, and the absorbance of the solution at 485 nm is monitored. What does the graph of A_{485} as a function of added volume of NaOH look like?



71. For the reaction



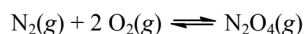
which statements are correct?

I. $K_c = [\text{SO}_2][\text{O}_2]/[\text{SO}_3]$

II. Addition of $\text{O}_2(\text{g})$ to the system at constant temperature and volume would decrease the value of K_c .

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

72. What is the value of K_p for the following reaction?



Reaction	K_p
$\frac{1}{2} \text{N}_2\text{O}_4(\text{g}) \rightleftharpoons \text{NO}_2(\text{g})$	x
$\frac{1}{2} \text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons \text{NO}_2(\text{g})$	y

- (A) y/x (B) y^2/x^2 (C) x^2/y^2 (D) xy^2

73. Which solution has the greatest percent ionization?

- (A) 0.10 M NH_3 ($K_b = 1.8 \times 10^{-5}$)
(B) 0.25 M HNO_2 ($K_a = 4.5 \times 10^{-4}$)
(C) 1.00 M HCOOH ($K_a = 1.7 \times 10^{-4}$)
(D) 2.00 M CH_3NH_2 ($K_b = 4.4 \times 10^{-4}$)

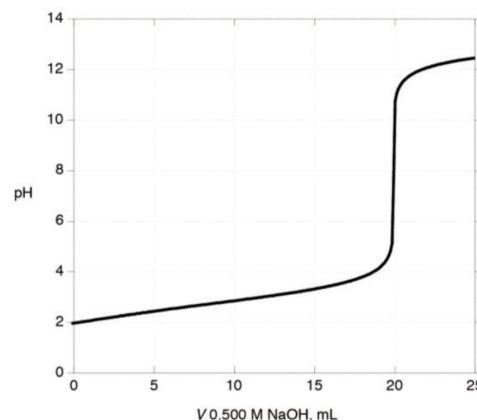
74. What is the pH of a 0.200 M solution of $\text{C}_6\text{H}_5\text{COONa}$? (The K_a of $\text{C}_6\text{H}_5\text{COOH}$ is 6.4×10^{-5} .)

- (A) 5.25 (B) 5.40 (C) 8.60 (D) 8.75

75. Silver chloride, AgCl ($K_{sp} = 1.8 \times 10^{-10}$), can be dissolved in solutions containing ammonia due to the formation of the soluble complex ion $\text{Ag}(\text{NH}_3)_2^+$ ($K_f = 1.0 \times 10^8$). What is the minimum amount of NH_3 that would need to be added to dissolve 0.010 mol AgCl in 1.00 L of solution?

- (A) 0.010 mol (B) 0.020 mol
(C) 0.095 mol (D) 0.13 mol

76. A sample of 100 mL of a solution of a weak monoprotic acid of unknown concentration is titrated with 0.500 M NaOH to give the titration curve shown.



All of the statements are correct EXCEPT:

- (A) Phenolphthalein would be a suitable indicator for this titration.
(B) A buffer solution is formed when 15 mL of NaOH is added.
(C) The pK_a of the acid is 4.0.
(D) The initial concentration of the acid is 0.10 M.

77. A 50.0 mL sample of a 1.00 M solution of a diprotic acid H_2A ($K_{a1} = 1.0 \times 10^{-6}$ and $K_{a2} = 1.0 \times 10^{-10}$) is titrated with 2.00 M NaOH. What is the minimum volume of 2.00 M NaOH needed to reach a pH of 10.00?

- (A) 12.5 mL (B) 37.5 mL
(C) 25.0 mL (D) 50.0 mL

78. For the reaction, $\text{O}_2(\text{g}) + 2 \text{F}_2(\text{g}) \rightleftharpoons 2 \text{OF}_2(\text{g})$, $K_p = 41.0$. If $P_{\text{O}_2}(\text{g}) = 0.116$ atm and $P_{\text{F}_2}(\text{g}) = 0.0461$ atm at equilibrium, what is the pressure of $\text{OF}_2(\text{g})$?

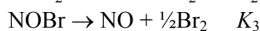
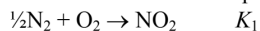
- (A) 0.101 atm (B) 0.132 atm
(C) 0.760 atm (D) 166 atm

Equilibrium

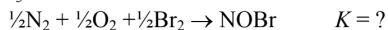
79. For the reaction $2 \text{HI(g)} \rightleftharpoons \text{H}_2\text{(g)} + \text{I}_2\text{(g)}$, what is the relationship between K_c and K_p at 25°C ?
- (A) $K_c = K_p$
 (B) $K_c > K_p$
 (C) $K_c < K_p$
 (D) The relationship varies depending on the pressure.
80. Calcium hydroxide is slightly soluble in water with a K_{sp} of 1.3×10^{-6} . What is the pH of a saturated solution of calcium hydroxide at 25°C ?
- (A) 12.34 (B) 12.14 (C) 12.04 (D) 11.84
81. For the reaction, $\text{ADP} + \text{phosphate} \rightarrow \text{ATP}$, $\Delta G^\circ = 30.50 \text{ kJ}\cdot\text{mol}^{-1}$. What is the value of the equilibrium constant, K , for this process under physiological conditions of 37.5°C ?
- (A) 4.5×10^{-6} (B) 7.4×10^{-6}
 (C) 1.3×10^5 (D) 2.2×10^5
82. When 0.0030 mol of HCl is added to 100. mL of a 0.10 M solution of a weak base, R_2NH , the solution has a pH of 11.10. What is K_b for the weak base?
- (A) 2.9×10^{-3} (B) 5.4×10^{-4}
 (C) 4.1×10^{-5} (D) 1.6×10^{-5}
83. For which reaction will K_p be larger than K_c at 25°C ?
- (A) $\text{CO}_2\text{(g)} + \text{C(s)} \rightleftharpoons 2 \text{CO(g)}$
 (B) $2 \text{NO}_2\text{(g)} \rightleftharpoons \text{N}_2\text{O}_4\text{(g)}$
 (C) $\text{H}_2\text{(g)} + \text{F}_2\text{(g)} \rightleftharpoons 2 \text{HF(g)}$
 (D) $\text{O}_3\text{(g)} + \text{NO(g)} \rightleftharpoons \text{NO}_2\text{(g)} + \text{O}_2\text{(g)}$
84. $2 \text{SO}_3\text{(g)} \rightleftharpoons 2 \text{SO}_2\text{(g)} + \text{O}_2\text{(g)}$
 When $\text{SO}_3\text{(g)}$ is added to a sealed bulb at a pressure of 2.0 atm, it undergoes the reaction above. At equilibrium, 76% of the $\text{SO}_3\text{(g)}$ has reacted. What is the value of K_p at this temperature?
- (A) 15 (B) 7.6 (C) 3.8 (D) 2.4
85. The pH of pure water at 50°C is 6.63. What is the value of K_w at 50°C ?
- (A) 1.8×10^{-15} (B) 1.0×10^{-14}
 (C) 5.5×10^{-14} (D) 2.2×10^{-13}
86. Alanine, $\text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$, has $K_a = 4.5 \times 10^{-3}$ and $K_b = 7.4 \times 10^{-5}$. Which species has the highest concentration at a pH of 7.00 in H_2O ?
- (A) $\text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$ (B) $^+\text{H}_3\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$
 (C) $\text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2^-$ (D) $^+\text{H}_3\text{NCH}(\text{CH}_3)\text{CO}_2^-$
87. What is the pH of the solution formed by mixing 25.0 mL of a 0.15 M solution of NH_3 with 25.0 mL of 0.12 M HCl? (K_b for $\text{NH}_3 = 1.8 \times 10^{-5}$)
- (A) 4.14 (B) 5.34 (C) 8.65 (D) 9.86
88. What is the $[\text{OH}^-]$ in a suspension of the antacid $\text{Mg}(\text{OH})_2$? ($K_{sp} = 2.06 \times 10^{-13}$)
- (A) $7.4 \times 10^{-5} \text{ M}$ (B) $5.9 \times 10^{-5} \text{ M}$
 (C) $4.7 \times 10^{-5} \text{ M}$ (D) $3.7 \times 10^{-5} \text{ M}$
89. A 1 M aqueous solution of which molecule has the lowest pH?
- (A) HOCl (B) H_2SO_3 (C) H_3PO_4 (D) H_2SO_4
90. If the initial pH values are the same for titrations of separate 25 mL samples of weak and strong monoprotic acids, which other value(s) is(are) also the same?
 I the pH at the equivalence point
 II the volume of base needed to reach the eq. point
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
91. The addition of 0.01 mol of which of the following to 100 mL of H_2O will give the most alkaline aqueous solution?
- (A) NH_3 (B) HONH_2
 (C) CH_3NH_2 (D) H_2NNH_2
92. What is the pH of a 1.00 L sample of a buffer solution containing 0.10 mol of benzoic acid and 0.10 mol of sodium benzoate to which 0.010 mol of NaOH has been added? [K_a benzoic acid = 6.5×10^{-5}]
- (A) 4.27 (B) 4.23
 (C) 4.15 (D) 4.10
93. Equal volumes of $1 \times 10^{-4} \text{ M}$ solutions of Cd^{2+} and CO_3^{2-} ions are mixed in one flask and equal volumes of $1 \times 10^{-4} \text{ M}$ solutions of Ag^+ and CrO_4^{2-} ions are mixed in a second. Which substances precipitate?
- | Formula | CdCO_3 | Ag_2CrO_4 |
|----------|-----------------------|---------------------------|
| K_{sp} | 5.2×10^{-12} | 1.1×10^{-12} |
- (A) CdCO_3 only (B) Ag_2CrO_4 only
 (C) Both (D) Neither

Equilibrium

94. Consider these reactions and their corresponding K s.



Express the K value for the reaction below in terms of K_1 , K_2 , and K_3 .

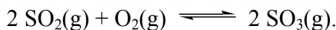


- (A) $K_1 + K_2/2 - K_3$ (B) $K_1 + (K_2)^{1/2} - K_3$
(C) $K_1 K_2 / 2 K_3$ (D) $K_1 (K_2)^{1/2} / K_3$

95. For the hypothetical reaction: $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$, the equilibrium constant, K , is less than 1.0 at 25 °C and decreases by 35% on changing the temperature to 45 °C. What must be true according to this information?

- (A) The ΔH° for the reaction is negative.
(B) The ΔS° for the reaction is positive.
(C) The ΔG° for the reaction at 25° is negative.
(D) The ΔG° for the reaction at 45° is zero.

96. Consider the reaction carried out at constant volume:



For initial concentrations of SO_2 and O_2 of 2.0 M and 1.5 M, respectively, the equilibrium O_2 concentration is 0.80 M. What is the value of K_c for this reaction?

- (A) 6.8 (B) 2.9 (C) 0.34 (D) 0.15

97. Correct statements about the percentage ionization of weak acids in water include which of the following?

- I The percentage ionization increases as the ionization constant of the acid becomes larger.
II The percentage ionization increases as the concentration of the acid becomes smaller.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

98. What is the pH of the solution made by mixing 25.0 mL of 1.00×10^{-3} M HNO_3 and 25.0 mL of 1.00×10^{-3} M NH_3 ? [K_b for $\text{NH}_3 = 1.8 \times 10^{-5}$]

- (A) 4.02 (B) 6.28 (C) 7.72 (D) 9.98

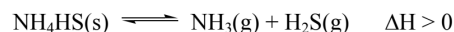
99. For the synthesis of ammonia, K_c is 1.2 at 375 °C
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$,
What is K_p at this temperature?

- (A) 4.1×10^{-8} (B) 4.2×10^{-4}
(C) 1.3×10^{-3} (D) 3.4×10^3

100. Calculate the aqueous solubility of $\text{Ca}(\text{OH})_2$ in grams per liter. [$K_{sp} = 8.0 \times 10^{-6}$]

- (A) 5.9×10^{-4} (B) 2.0×10^{-2}
(C) 0.93 (D) 1.5

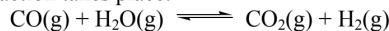
101. Consider the system at equilibrium:



Factors which favor the formation of more $\text{H}_2\text{S}(\text{g})$ include which of the following?

- I adding a small amount of $\text{NH}_4\text{HS}(\text{s})$ at constant volume
II increasing the pressure at constant temperature
III increasing the temperature at constant pressure
(A) I only (B) III only
(C) I and II only (D) I and III only

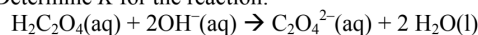
102. A 2.0 L container is charged with a mixture of 6.0 moles of $\text{CO}(\text{g})$ and 6.0 moles of $\text{H}_2\text{O}(\text{g})$ and the following reaction takes place:



When equilibrium is reached the $[\text{CO}_2] = 2.4$ M. What is the value of K_c for the reaction?

- (A) 16 (B) 4.0 (C) 0.25 (D) 0.063

103. Determine K for the reaction:



$\text{H}_2\text{C}_2\text{O}_4(\text{aq})$	$K_{a1} = 6.5 \times 10^{-2}$	$K_{a2} = 6.1 \times 10^{-5}$
H_2O	$K_w = 1.0 \times 10^{-14}$	

- (A) 4.0×10^{-34} (B) 4.0×10^{-6}
(C) 4.0×10^6 (D) 4.0×10^{22}

104. Which range includes the value of the equilibrium constant, K_{eq} , for a system with $\Delta G^\circ \ll 0$?

- (A) $-1 < K_{eq} < 0$ (B) $0 < K_{eq} < 1$
(C) $K_{eq} < -1$ (D) $1 < K_{eq}$

105. What volumes of 0.200 M HNO_2 and 0.200 M NaNO_2 are required to make 500. mL of a buffer solution with pH = 3.00? [K_a for $\text{HNO}_2 = 4.00 \times 10^{-4}$]

- (A) 250. mL of each
(B) 143 mL of HNO_2 and 357 mL of NaNO_2
(C) 200. mL of HNO_2 and 300. mL of NaNO_2
(D) 357 mL of HNO_2 and 143 mL of NaNO_2

106. A sample of sparingly soluble $\text{PbI}_2(\text{s})$ containing radioactive I-133 is added to 0.10 M $\text{KI}(\text{aq})$ and stirred overnight. Observations about this system include which of the following?

- I The radioactivity of the liquid phase increases significantly.
II The concentration of the I^- ion in solution increases significantly.
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Equilibrium

107. The equilibrium system, $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, has $K_p = 11$. For which equilibrium system is $K_p = 0.091$?
- (A) $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ (B) $\text{NO}_2(\text{g}) \rightleftharpoons 1/2\text{N}_2\text{O}_4(\text{g})$
- (C) $2\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 4\text{NO}_2(\text{g})$ (D) $1/2\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons \text{NO}_2(\text{g})$

108. A 1.0 L evacuated tank is charged with HI(g) to a pressure of 1.0 atm at 793K. Some of the HI(g) forms $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ according to the equilibrium;
- $$2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g}) \quad K_p = 0.016$$
- What is the pressure (in atm) of HI at equilibrium?
- (A) 0.11 (B) 0.13 (C) 0.80 (D) 1.6

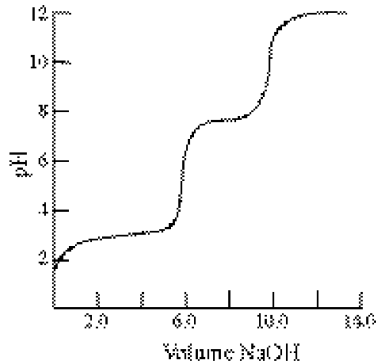
109. A solution of 0.10 M NaZ has a pH = 8.90. What is the K_a of HZ?
- (A) 1.6×10^{-4} (B) 1.6×10^{-5}
- (C) 6.3×10^{-10} (D) 6.3×10^{-11}

110. What is the pH of a solution made by adding 0.41g of $\text{NaC}_2\text{H}_3\text{O}_2$ to 100. mL of 0.10 M $\text{HC}_2\text{H}_3\text{O}_2$?

Molar mass, $\text{g} \cdot \text{mol}^{-1}$	
$\text{NaC}_2\text{H}_3\text{O}_2$	82.0
K_a	
$\text{HC}_2\text{H}_3\text{O}_2$	1.8×10^{-5}

- (A) 4.44 (B) 4.70 (C) 5.05 (D) 8.95

111. The curve represents the titration of



- (A) a diprotic acid.
- (B) two monoprotic acids with the same K_a s but different concentrations.
- (C) two monoprotic acids with different K_a s but the same concentrations.
- (D) two monoprotic acids with different K_a s and different concentrations.

112. Which silver compound is the most soluble in water?

- (A) AgCl $K_{sp} = 1.8 \times 10^{-10}$
- (B) Ag_2CO_3 $K_{sp} = 8.5 \times 10^{-12}$
- (C) AgBr $K_{sp} = 5.4 \times 10^{-13}$
- (D) Ag_3PO_4 $K_{sp} = 8.9 \times 10^{-17}$

113. When 2.00 mol each of $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ are reacted in a 1.00 L container at a certain temperature, 3.50 mol of HI is present at equilibrium. Calculate the value of the equilibrium constant, K_c .

- (A) 3.7 (B) 14 (C) 56 (D) 2.0×10^2

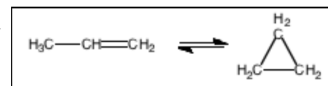
114. For which equation is the equilibrium constant equal to K_a for the ammonium ion, NH_4^+ ?

- (A) $\text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- (B) $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- (C) $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- (D) $\text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\text{l})$

115. What is the pH of a solution prepared by mixing 45.0 mL of 0.184 M KOH with 65.0 mL of 0.145 M HCl?

- (A) 1.07 (B) 1.13 (C) 1.98 (D) 2.92

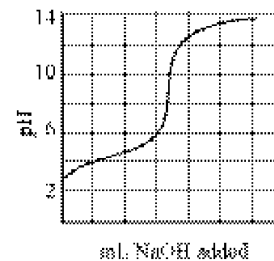
116. The gas phase reaction shown is endothermic as written. Which change(s) will increase the quantity of $\text{CH}_3\text{CH}=\text{CH}_2$ at equilibrium?



- I. increasing the temperature
II. increasing the pressure

- (A) I only (B) II only
- (C) Both I and II (D) Neither I nor II

117. The curve represents the titration of a weak monoprotic acid. Over what pH range(s) will the acid being titrated serve as a buffer when mixed with its salt?



- I. pH 4 – 6
II. pH 7 – 9
III. pH 12 – 13

- (A) I only (B) II only
- (C) I and III only (D) I, II and III

118. The pH of a saturated solution of $\text{Fe}(\text{OH})_2$ is 8.67. What is the K_{sp} for $\text{Fe}(\text{OH})_2$?

- (A) 5×10^{-6} (B) 2×10^{-11}
- (C) 1×10^{-16} (D) 5×10^{-17}

119. A 0.050 M solution of an unknown acid is 1.0% ionized. What is the value of its K_a ?

- (A) 2.5×10^{-7} (B) 5.0×10^{-6}
- (C) 5.0×10^{-4} (D) 5.0×10^{-2}

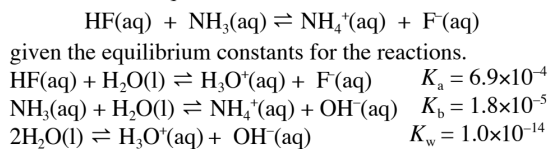
Equilibrium

120. Which mixture(s) form(s) buffer solutions?

- | | | |
|------|---|--|
| I. | 100 mL of 0.200 M HF | and 200 mL of 0.200 M NaF |
| II. | 200 mL of 0.200 M HCl | and 200 mL of 0.400 M CH ₃ CO ₂ Na |
| III. | 300 mL of 0.100 M CH ₃ CO ₂ H | and 100 mL of 0.300 M CH ₃ CO ₂ Na |

- (A) I only (B) III only
(C) II and III only (D) I, II and III

121. Determine the equilibrium constant for the reaction:



- (A) 1.2×10^{-8} (B) 1.2×10^6
(C) 8.1×10^7 (D) 3.8×10^{15}

122. Calculate the pH of a 0.15 M solution of HOCl.

K_a	
HOCl	2.9×10^{-8}

- (A) 3.77 (B) 4.18 (C) 6.71 (D) 8.36

123. For which reaction does $K_p = K_c$?

- (A) $2\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO(g)}$
(B) $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons 2\text{NH}_3\text{(g)}$
(C) $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{H}_2\text{O(g)}$
(D) $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightleftharpoons 2\text{HI(g)}$

124. CaF₂ has a $K_{sp} = 3.9 \times 10^{-11}$ at 25°C. What is the [F⁻] in a saturated solution of CaF₂ at 25°C?

- (A) 2.1×10^{-4} (B) 3.4×10^{-4}
(C) 4.3×10^{-4} (D) 6.8×10^{-4}

125. C(s) + CO₂(g) $\rightleftharpoons$ 2CO(g)

If this system is at equilibrium, which change(s) will alter the value of K_p ?

- | | |
|------|-------------------------|
| I. | raising the temperature |
| II. | adding solid C |
| III. | decreasing the pressure |

- (A) I only (B) II only
(C) I and III only (D) II and III only

126. A 0.10 M solution of a weak acid is 5.75% ionized. What is the K_a value for this acid?

- (A) 3.3×10^{-3} (B) 3.5×10^{-4}
(C) 4.2×10^{-5} (D) 3.3×10^{-5}

127. Which base is most suitable to prepare a buffer solution with a pH = 11.00?

- (A) ammonia ($K_b = 1.8 \times 10^{-5}$)
(B) aniline ($K_b = 4.0 \times 10^{-10}$)
(C) methylamine ($K_b = 4.4 \times 10^{-4}$)
(D) pyridine ($K_b = 1.7 \times 10^{-9}$)

128. Calculate the pH of a 0.10 M solution of H₂CO₃.

H ₂ CO ₃ Acid Ionization Constants	
K_{1a}	4.4×10^{-7}
K_{2a}	4.7×10^{-11}

- (A) 3.68 (B) 5.76 (C) 7.36 (D) 9.34

129. Which saturated solution has the highest [OH⁻]?

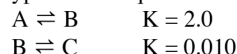
- (A) aluminum hydroxide ($K_{sp} = 1.8 \times 10^{-32}$)
(B) calcium hydroxide ($K_{sp} = 8.0 \times 10^{-6}$)
(C) iron(II) hydroxide ($K_{sp} = 1.6 \times 10^{-14}$)
(D) magnesium hydroxide ($K_{sp} = 1.2 \times 10^{-11}$)

130. Consider these mixtures:

Mixture I.	100 mL of 0.006 M Pb(NO ₃) ₂ plus 50 mL of 0.003 M NaBr
Mixture II.	100 mL of 0.008 M Pb(NO ₃) ₂ plus 100 mL of 0.006 M NaBr
Which statement is correct?	
	K_{sp}
	PbBr ₂ 6.6×10^{-6}

- (A) A precipitate will not form in either mixture.
(B) A precipitate will form only in mixture I.
(C) A precipitate will form only in mixture II.
(D) A precipitate will form in both mixtures.

131. For the hypothetical equilibrium reactions;



What is the value of K for the reaction;
 $2\text{C} \rightleftharpoons 2\text{A}$?

- (A) 2500 (B) 50 (C) 25 (D) 4.0×10^{-4}

132. For which reaction is $K_p = K_c$?

- | | |
|------|--|
| I. | $2\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{N}_2\text{O(g)}$ |
| II. | $\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons \text{CO}_2\text{(g)}$ |
| III. | $\text{N}_2\text{O}_4\text{(g)} \rightleftharpoons 2\text{NO}_2\text{(g)}$ |

- (A) II only (B) III only
(C) I and III only (D) II and III only

133. What is the pH of a 0.010 M solution of a weak acid HA that is 4.0% ionized?

- (A) 0.60 (B) 0.80 (C) 2.80 (D) 3.40

Equilibrium

134. Given the acid ionization constants, when the conjugate bases are arranged in order of increasing base strength, which order is correct?
- | Acid Ionization Constant, K_a | |
|---|-----------------------|
| HClO | 3.5×10^{-8} |
| HClO ₂ | 1.2×10^{-2} |
| HCN | 6.2×10^{-10} |
| H ₂ PO ₄ ⁻ | 6.2×10^{-8} |

- (A) ClO₂⁻, ClO⁻, HPO₄²⁻, CN⁻
 (B) ClO₂⁻, HPO₄²⁻, ClO⁻, CN⁻
 (C) CN⁻, HPO₄²⁻, ClO⁻, ClO₂⁻
 (D) CN⁻, ClO⁻, HPO₄²⁻, ClO₂⁻

135. Calculate the concentration of hydrogen ion in mol/L of a 0.010 M solution of NH₄Cl.
- | Base Ionization Constant, K_b | |
|---------------------------------|----------------------|
| NH ₃ | 1.8×10^{-5} |

- (A) 4.2×10^{-4} (B) 2.4×10^{-6}
 (C) 1.8×10^{-7} (D) 5.6×10^{-12}

136. For the reaction;
 $\text{PbI}_2(\text{s}) \rightleftharpoons \text{Pb}^{2+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) \quad K_{\text{sp}} = 8.4 \times 10^{-9}$
 What is the concentration of Pb²⁺ in mol/L in a saturated solution of PbI₂ in which [I⁻] = 0.01 M?

- (A) 8.4×10^{-7} (B) 8.4×10^{-5}
 (C) 1.3×10^{-3} (D) 2.0×10^{-3}

137. $\text{H}_2\text{S}(\text{aq}) \rightleftharpoons \text{H}^{+}(\text{aq}) + \text{HS}^{-}(\text{aq}) \quad K = 9.5 \times 10^{-8}$
 $\text{HS}^{-}(\text{aq}) \rightleftharpoons \text{H}^{+}(\text{aq}) + \text{S}^{2-}(\text{aq}) \quad K = 1.0 \times 10^{-19}$
 Given the equilibrium constants provided, what is the equilibrium constant for the reaction;
 $\text{S}^{2-}(\text{aq}) + 2\text{H}^{+}(\text{aq}) \rightleftharpoons \text{H}_2\text{S}(\text{aq}) \quad K = ?$

- (A) 9.5×10^{-27} (B) 9.7×10^{-14}
 (C) 9.5×10^{11} (D) 1.0×10^{26}

138. Calculate the hydronium ion concentration in 50.0 mL of 0.10 M NaH₂AsO₄.
 ($K_1 = 6.0 \times 10^{-3}$, $K_2 = 1.1 \times 10^{-7}$, $K_3 = 3.0 \times 10^{-12}$)

- (A) 2.4×10^{-2} (B) 1.6×10^{-3}
 (C) 1.0×10^{-4} (D) 2.5×10^{-5}

139. When the acids; HClO₃, H₃BO₃, H₃PO₄, are arranged in order of increasing strength, which order is correct?

- (A) H₃BO₃ < H₃PO₄ < HClO₃
 (B) HClO₃ < H₃BO₃ < H₃PO₄
 (C) H₃PO₄ < HClO₃ < H₃BO₃
 (D) H₃BO₃ < HClO₃ < H₃PO₄

140. A buffer solution results from mixing equal volumes of which solutions?

- I 0.10 M HCl and 0.20 M NH₃
 II 0.10 M HNO₂ and 0.10 M NaNO₂
 III 0.20 M HCl and 0.10 M NaCl

- (A) II only (B) I and II only
 (C) I and III only (D) I, II and III

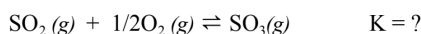
141. A solution is 0.10 M in Ag⁺, Ca²⁺, Mg²⁺, and Al³⁺ ions. Which compound will precipitate at the lowest [PO₄³⁻] when a solution of Na₃PO₄ is added?

- (A) Ag₃PO₄ ($K_{\text{sp}} = 1 \times 10^{-16}$)
 (B) Ca₃(PO₄)₂ ($K_{\text{sp}} = 1 \times 10^{-33}$)
 (C) Mg₃(PO₄)₂ ($K_{\text{sp}} = 1 \times 10^{-24}$)
 (D) AlPO₄ ($K_{\text{sp}} = 1 \times 10^{-20}$)

142. Which salt is significantly more soluble in a strong acid than in water?

- (A) PbF₂ (B) PbCl₂ (C) PbBr₂ (D) PbI₂

143. Based on the equilibrium constant for the reaction below,
 $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \quad K = 1.8 \times 10^{-5}$
 what is the equilibrium constant for the reaction



- (A) 2.1×10^{-3} (B) 4.2×10^{-3}
 (C) 2.4×10^2 (D) 5.6×10^4

144. Calculate the [H⁺] in a 0.25 M solution of methylamine, CH₃NH₂ ($K_b = 4.4 \times 10^{-4}$).

- (A) 1.1×10^{-4} (B) 1.0×10^{-2}
 (C) 9.1×10^{-11} (D) 9.5×10^{-13}

145. A 0.010 M solution of a weak acid, HA, is 0.40% ionized. What is its ionization constant?

- (A) 1.6×10^{-10} (B) 1.6×10^{-7}
 (C) 4.0×10^{-5} (D) 4.0×10^{-3}

146. 1.0 L of an aqueous solution in which

[H₂CO₃] = [HCO₃⁻] = 0.10 M, has [H⁺] = 4.2×10^{-7} . What is the [H⁺] after 0.005 mole of NaOH has been added?

- (A) 2.1×10^{-9} M (B) 2.2×10^{-8} M
 (C) 3.8×10^{-7} M (D) 4.6×10^{-7} M

Equilibrium

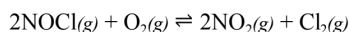
147. A solution of $\text{Pb}(\text{NO}_3)_2$ is added dropwise to a second solution in which $[\text{Cl}^-] = [\text{F}^-] = [\text{I}^-] = [\text{SO}_4^{2-}] = 0.001 \text{ M}$. What is the first precipitate that forms?

- (A) PbCl_2 ($K_{\text{sp}} = 1.5 \times 10^{-5}$)
 (B) PbF_2 ($K_{\text{sp}} = 3.7 \times 10^{-8}$)
 (C) PbI_2 ($K_{\text{sp}} = 8.5 \times 10^{-9}$)
 (D) PbSO_4 ($K_{\text{sp}} = 1.8 \times 10^{-8}$)

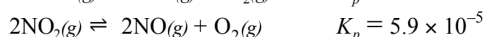
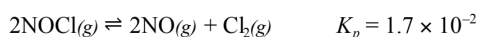
148. A 0.1 M solution of which substance is most acidic?

- (A) NaHSO_4 (B) Na_2SO_4
 (C) NaHS (D) NaHCO_3

149. Determine the value of the equilibrium constant for this reaction



from the K values for these reactions.



- (A) 1.0×10^{-6} (B) 1.0×10^{-3}
 (C) 3.5×10^{-3} (D) 2.9×10^2

150. What is the pH of a 0.15 M solution of hydrazine, N_2H_4 ?

Hydrazine	K_b
N_2H_4	1.0×10^{-6}

- (A) 3.41 (B) 6.82 (C) 10.59 (D) 11.00

Questions 151 and 152 refer to aqueous solutions of formic acid, HCOOH , which has a K_a value of 1.9×10^{-4} at 25°C .

151. What is the percent ionization of a 0.10 M solution of formic acid at 25°C ?

- (A) 0.19% (B) 1.4% (C) 4.4% (D) 14%

152. How many moles of sodium formate must be added to 1.0 L of a 0.20 M formic acid solution to produce a pH of 4.00?

- (A) 0.38 (B) 0.80 (C) 1.9 (D) 3.8

153. During the titration of a weak base with a strong acid, one should use an acid-base indicator that changes color in the

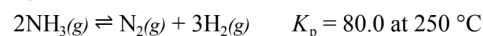
- (A) acidic range. (B) basic range.
 (C) buffer range. (D) neutral range.

154. What is the solubility of calcium hydroxide in $\text{mol}\cdot\text{L}^{-1}$?

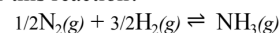
Substance	K_{sp}
calcium hydroxide	4.0×10^{-6}

- (A) 1.6×10^{-2} (B) 1.0×10^{-2}
 (C) 2.0×10^{-3} (D) 1.0×10^{-3}

Questions 155 and 156 should both be answered with reference to this equilibrium system.

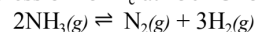


155. What is K_p for this reaction?



- (A) 0.0125 (B) 0.112
 (C) 8.94 (D) 40.0

156. What is the expression for K_c at 250°C for this reaction?



- (A) $K_c = \frac{K_p}{(RT)^2}$ (B) $K_c = \frac{K_p}{RT}$
 (C) $K_c = K_p(RT)^2$ (D) $K_c = K_p RT$

157. $\text{HCOOH}(aq) \rightleftharpoons \text{H}^+(aq) + \text{HCOO}^-(aq)$ $K_a = 1.7 \times 10^{-4}$

The ionization of formic acid is represented. Calculate $[\text{H}^+]$ of a solution initially containing 0.10 M HCOOH and 0.050 M HCOONa .

- (A) $8.5 \times 10^{-5} \text{ M}$ (B) $3.4 \times 10^{-4} \text{ M}$
 (C) $4.1 \times 10^{-3} \text{ M}$ (D) $1.8 \times 10^{-2} \text{ M}$

158. Which are strong acids?

I. HClO_3 II. H_2SeO_3 III. H_3AsO_4

- (A) I only (B) III only
 (C) I and III only (D) II and III only

159. Carbonic acid, H_2CO_3 , is a diprotic acid for which $K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.7 \times 10^{-11}$. Which solution will produce a pH closest to 9?

- (A) 0.1 M H_2CO_3 (B) 0.1 M Na_2CO_3
 (C) 0.1 M NaHCO_3 (D) 0.1 M NaHCO_3 and 0.1 M Na_2CO_3

160. What is the pH of a saturated solution of magnesium hydroxide, $\text{Mg}(\text{OH})_2$ at 25°C ? ($K_{\text{sp}} = 6.0 \times 10^{-12}$ at 25°C)

- (A) 10.56 (B) 10.36
 (C) 10.26 (D) 10.05

Equilibrium

161. For the reaction $2A(g) + 2B(g) \rightleftharpoons 3C(g)$ at a certain temperature, K is 2.5×10^{-2} . For which conditions will the reaction proceed to the right at the same temperature?

	[A], M	[B], M	[C], M
(A)	0.10	0.10	0.10
(B)	1.0	1.0	1.0
(C)	1.0	0.10	0.10
(D)	1.0	1.0	0.10

162. What is the K_b of a weak base that produces one OH^- per molecule if a 0.050 M solution is 2.5% ionized?

- (A) 7.8×10^{-8} (B) 1.6×10^{-6}
(C) 3.2×10^{-5} (D) 1.2×10^{-3}

163. What is the $[OH^-]$ of a 0.65 M solution of NaOCl?

Acid	K_a
HOCl	2.8×10^{-8}

- (A) 4.8×10^{-4} M (B) 1.3×10^{-4} M
(C) 3.5×10^{-7} M (D) 2.1×10^{-11} M

164. Which acid is the strongest?

- (A) H_3BO_3 (B) H_3PO_4
(C) H_2SO_3 (D) $HClO_3$

165. What is the conjugate acid of HPO_4^{2-} ?

- (A) $H_3PO_4(aq)$ (B) $H_2PO_4^-(aq)$
(C) $H_3O^+(aq)$ (D) $PO_4^{3-}(aq)$

166. The amount of sodium hydrogen carbonate, $NaHCO_3$, in an antacid tablet is to be determined

Acid	K_a
H_2CO_3	2.5×10^{-4}
HCO_3^-	2.4×10^{-8}

by dissolving the tablet in water and titrating the resulting solution with hydrochloric acid. Which indicator is the most appropriate for this titration?

- (A) methyl orange, $pK_{in} = 3.7$
(B) bromothymol blue, $pK_{in} = 7.0$
(C) phenolphthalein, $pK_{in} = 9.3$
(D) alizarin yellow, $pK_{in} = 12.5$

167. How many moles of NaOCl must be added to 150 mL of 0.025 M HOCl to obtain a buffer solution with a pH = 7.50?

Acid	K_a
HOCl	2.8×10^{-8}

- (A) 2.6×10^{-5} (B) 1.1×10^{-3}
(C) 3.3×10^{-3} (D) 2.2×10^{-2}

168. If equal volumes of $BaCl_2$ and NaF solutions are mixed, which of these combinations will **not** give a precipitate?

Substance	K_{sp}
BaF_2	1.7×10^{-7}

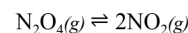
- (A) 0.0040 M $BaCl_2$ and 0.020 M NaF
(B) 0.010 M $BaCl_2$ and 0.015 M NaF
(C) 0.015 M $BaCl_2$ and 0.010 M NaF
(D) 0.020 M $BaCl_2$ and 0.0020 M NaF

169. $P_4(s) + 6Cl_2(g) \rightleftharpoons 4PCl_3(g)$

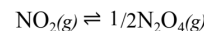
Phosphorus reacts with chlorine as shown. What is the equilibrium constant expression, K_p , for this reaction?

- (A) $\frac{4P_{PCl_3}}{6P_{PCl_3} \cdot P_{Cl_2}}$ (B) $\frac{4P_{PCl_3}}{6P_{Cl_2}}$
(C) $\frac{P_{PCl_3}}{P_{P_4} \cdot P_{Cl_2}^6}$ (D) $\frac{P_{PCl_3}^4}{P_{Cl_2}^6}$

170. The equilibrium constant for the reaction



is 6.10×10^{-3} at $25^\circ C$. Calculate the value of K for this reaction:

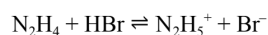
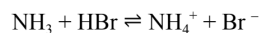
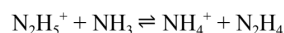


- (A) 327 (B) 164
(C) 12.8 (D) 3.05×10^{-3}

171. The ion-product constant for water at $45^\circ C$ is 4.0×10^{-14} . What is the pH of pure water at this temperature?

- (A) 6.7 (B) 7.0 (C) 7.3 (D) 13.4

172. The position of equilibrium lies to the right in each of these reactions.



Based on this information, what is the order of acid strength?

- (A) $HBr > N_2H_5^+ > NH_4^+$ (B) $N_2H_5^+ > N_2H_4 > NH_4^+$
(C) $NH_3 > N_2H_4 > Br^-$ (D) $N_2H_5^+ > HBr > NH_4^+$

173. HCN is a weak acid ($K_a = 6.2 \times 10^{-10}$). NH_3 is a weak base ($K_b = 1.8 \times 10^{-5}$). A 1.0 M solution of NH_4CN would be

- (A) strongly acidic (B) weakly acidic
(C) neutral (D) weakly basic

Equilibrium

174. What is the percent ionization of a 0.010 M HCN solution? ($K_a = 6.2 \times 10^{-10}$)
- (A) 0.0025% (B) 0.025%
(C) 0.25% (D) 2.5%
175. How many moles of HCOONa must be added to 1.0 L of 0.10 M HCOOH to prepare a buffer solution with a pH of 3.4? (HCOOH $K_a = 2 \times 10^{-4}$)
- (A) 0.01 (B) 0.05 (C) 0.1 (D) 0.2
176. The acid–base indicator methyl red has a K_a of 1×10^{-5} . Its acidic form is red while its alkaline form is yellow. If methyl red is added to a colorless solution with a pH = 7, the color will be
- (A) pink (B) red (C) orange (D) yellow
177. Silver ions are added to a solution with $[\text{Br}^-] = [\text{Cl}^-] = [\text{CO}_3^{2-}] = [\text{AsO}_4^{3-}] = 0.1 \text{ M}$. Which compound will precipitate at the lowest $[\text{Ag}^+]$?
- (A) AgBr ($K_{sp} = 5.0 \times 10^{-13}$)
(B) AgCl ($K_{sp} = 1.8 \times 10^{-10}$)
(C) Ag_2CO_3 ($K_{sp} = 8.1 \times 10^{-12}$)
(D) Ag_3AsO_4 ($K_{sp} = 1.0 \times 10^{-22}$)